HOFFMANN LABORATORY

Cow Uterine Contraction Analysis

Simple user guide for installing the repository, processing data, and regenerating figures

Repository: github.com/Yashwitha-7/Cow-Uterine-Contractions-Analysis
Audience: Hoffmann Laboratory collaborators
Prepared: August 2026

1. What this repository does

The repository provides a local web application for ingesting cow contraction TXT files and optional bolus Excel files, checking data quality, reviewing possible strain-polarity reversals, detecting exploratory contraction candidates, calculating day/night and 24-hour statistics, and generating figures.

Scientific caution: Detected peaks are candidate strain events. They are not confirmed physiological uterine contractions.

2. Before you begin

- macOS, Linux, or Windows with a terminal application
- Git
- Python 3.12 recommended
- Node.js 20 or later and npm
- Raw contraction TXT files and, when available, the bolus Excel file

Data location: Research data are deliberately excluded from GitHub. Obtain the raw data separately from the laboratory shared folder and never commit cow data to the public repository.

3. Download the repository

```
git clone https://github.com/Yashwitha-7/Cow-Uterine-Contractions-Analysis.git
cd Cow-Uterine-Contractions-Analysis
```

4. Install the backend

```
cd backend
python3.12 -m venv .venv
source .venv/bin/activate
pip install --upgrade pip
pip install -r requirements.txt
```

Windows activation command:

```
.venv\Scripts\activate
```

5. Install the frontend

```
cd ../frontend
npm install
```

6. Start the application

Terminal 1 — backend

```
cd Cow-Uterine-Contractions-Analysis/backend
source .venv/bin/activate
uvicorn app.main:app --reload
```

Terminal 2 — frontend

```
cd Cow-Uterine-Contractions-Analysis/frontend
npm run dev
```

Open the local address displayed by Vite, normally http://localhost:5173. The API documentation is available at http://127.0.0.1:8000/docs.

7. Recommended analysis sequence

1. Open Upload Data. Enter the cow ID and known calving date and time.
2. Select Contractions TXT files and upload all TXT files for that cow in one batch.
3. If available, upload one bolus Excel file for the same cow.
4. Open QC Logs and Data Preview to confirm that files were read correctly.
5. Open Polarity Review, enter the cow ID, and select Screen polarity.
6. Review every flagged section using the raw strain, centered strain, movement channels, zero axis, time axis, and 30-minute context.
7. Choose Keep polarity, Flip section, or Uncertain / exclude for every section.
8. Use the hourly signal browser to inspect any hour manually.
9. Open Phase 3 Processing and select Run Reviewed Analysis.
10. Open Visual Analysis or Downloads to view the saved figures, statistics, and CSV files.

8. How to make polarity decisions

Decision	Use when
Keep	The flagged interval is consistent with surrounding signal direction.
Flip	The complete flagged interval is clearly reversed relative to the surrounding recording.
Uncertain / exclude	The signal direction cannot be established confidently.

Important: Only the reviewed interval is flipped. The 30-minute context is displayed for comparison and is not modified.

9. Movement interpretation

The contraction files contain a binary movement flag derived from the accelerometer and gyroscope channels:

- 0 means no movement detected.
- 5 means movement detected.
- No intermediate values occurred in the available files.

The analysis also calculates acceleration and gyroscope variability independently. A strain peak near device-flagged or calculated movement is labeled movement-associated.

10. Where outputs are saved

```
data/
  raw/cow_<id>/contractions/
  raw/cow_<id>/bolus/
  processed/cow_<id>/
```

```
    quality_control/
    statistics/
    figures/
    clocklab_exports/
  database/hoffmann_lab.db
```

- quality_control contains polarity screening, review manifests, decisions, and archived decisions.
- statistics contains day/night CSV files and rhythm-summary JSON files.
- figures contains regenerated PNG visualizations.
- clocklab_exports contains CSV and AWD-formatted exports.

11. Reproducing the existing figures

To reproduce the same figures, collaborators need the same raw files, cow ID, calving timestamp, clock offsets, polarity decisions, software version, and analysis configuration. Recommended sharing package:

- Raw contraction and bolus files in the laboratory shared folder
- The relevant Git commit identifier
- The polarity-decisions CSV
- Cow metadata, including calving date and time
- The generated statistics, figures, and processed CSV files

Reproducibility: Do not compare regenerated outputs unless the raw files, review decisions, and code version are the same.

12. Verification commands

```
cd backend
.venv/bin/python -m pytest tests -q

cd ../frontend
npm run lint
npm run build
```

13. Common problems

Problem	Recommended action
No module named app	Run pytest from the backend folder, or use the documented command.
API returns Not Found	Confirm the URL begins with /api and that the backend is running.
Analysis is locked	Complete all pending polarity decisions first.
No figures appear	Run Reviewed Analysis, then reload Visual Analysis or Downloads.
Duplicate upload blocked	The cow already has that data type stored. Do not delete data unless the intended dataset is confirmed.
Port already in use	Stop the earlier backend or frontend process before starting another.

14. Safe data-handling rules

- Do not modify files under data/raw.
- Do not upload cow data, the SQLite database, or generated CSVs to the public GitHub repository.
- Back up the data folder before resetting or replacing a cow dataset.
- Record the code commit used for each sponsor-facing result package.
- Describe clean peaks as contraction candidates, not confirmed contractions.